What Makes a County Bounce Back?

Predicting Economic Recovery Across Two Very Different Crises

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2026-07-30

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1 Abstract

When a recession hits, some counties climb back to where they were within a few years. Others never do. We wanted to know whether that difference is predictable — and specifically, whether the traits that helped a county recover from the 2008 financial crisis also helped it recover from the COVID-19 downturn.

We built a dataset covering about 3,000 U.S. counties and measured recovery as employment two to three years after each crisis, compared to the peak before it. We then tested four pre-crisis traits: self-employment, migration, unemployment, and the share of the population that is working age. We controlled for wealth throughout, because we did not want to rediscover the obvious fact that richer places recover more easily.

Two traits predicted recovery in both crises. One predicted recovery only from COVID. One predicted nothing. When we added industry data for a smaller set of counties, we found that construction-heavy counties struggled in 2008 and were completely unaffected in COVID, while information-sector counties showed the opposite pattern.

Our most surprising result came from a test we ran to check our own work. We tried to predict employment change over a calm, non-crisis period and found it was about as predictable as COVID recovery. That means COVID was ordinary. The 2008 financial crisis, by contrast, was only half as predictable as a normal year — the real anomaly in our data.

2 Introduction

Every recession leaves some places behind. Drive through parts of Ohio or Michigan and you can still see the 2008 financial crisis in boarded-up storefronts. Drive through parts of Nevada or Florida and you will see the same thing from 2020. But the two sets of places are not the same places, and that is the puzzle that started this project.

The 2008 crisis began in housing and credit markets. The 2020 downturn began with a virus and the shutdowns that followed. They had almost nothing in common except that both wrecked local economies. If the same kinds of counties bounced back from both, that would suggest economic resilience is a durable property of a place — something baked into how a local economy is built. If different counties bounced back from each, resilience is not a property of the place at all. It depends on how the shock arrives.

That distinction matters beyond academic curiosity. Federal recovery aid is allocated using formulas built on county characteristics. If the traits that predict vulnerability change from crisis to crisis, a single formula cannot work for both.

2.1 The wealth problem

There is one obvious answer to "which counties recover best," and it is boring: the rich ones. Wealthier counties have more savings, more diversified employers, better credit access, and more capacity to absorb a shock. This is well documented and nobody needs another study to confirm it.

So we designed the project specifically to look past wealth. Our question is not "do rich counties recover better." It is: **among counties that were similarly well-off before a crisis, what separated the ones that bounced back from the ones that stayed stuck?**

3 Research Question

> Among counties with similar pre-crisis wealth, which measurable traits best predict how fully a county recovers from an economic crisis — and are those traits the same for the 2008 financial crisis and the COVID-19 downturn?

This is an **associational** study. We can identify which traits move together with recovery. We cannot prove that any trait *causes* recovery, because we are observing counties as they exist rather than running an experiment.

4 Data

4.1 Sources

We pulled from six federal sources and combined them into a single file.

Table 1: Primary data sources

Source	What we used it for
BLS Local Area Unemployment Statistics (LAUS)	County employment, used to build the recovery measure; pre-crisis unemployment rate
Census SAIPE	Median household income and poverty rate
Census Population Estimates Program (PEP)	Total population, working-age share, population trend
Census land area	Denominator for population density
IRS Statistics of Income	County-to-county migration flows
Bureau of Economic Analysis (BEA)	Proprietor share, our self-employment measure
USDA Economic Research Service	Rural-Urban Continuum Code

An unusual one worth explaining: the IRS migration data. When people move, they file taxes from a new address, which lets the IRS count roughly how many people entered and left each county. It is an odd source for an economics project, but it is the best county-level migration data that exists.

4.2 Measuring recovery

Our outcome variable is a ratio:

$$\text{recovery ratio} = \frac{\text{employment 2–3 years after the trough}}{\text{peak employment before the crisis}}$$

A value of 1.0 means a county got back exactly to where it started. Above 1.0 means it recovered and grew. Below 1.0 means it was still down.

We found each county's own peak and trough inside a search window rather than using fixed national dates, because crises did not hit every county on the same schedule. For the 2008 crisis, the peak is a county's highest employment year in 2006–2008, and the trough is its lowest employment year from the year after its peak through 2013. For COVID, the peak window is 2018–2019 and the trough window runs through 2022.

The numerator is then the average of employment in that county's trough year plus two and trough year plus three. Every county gets its own peak year, trough year, and measurement years, but the plus-two/plus-three offset rule is identical across both crises — which is what keeps the comparison fair.

We kept every county, including those that never recovered. They show up as low values, and there are a lot of them — a real feature of the data, not a problem to clean away.

A note on the transform. We take the natural log of the recovery ratio in every model. The raw ratio is bounded below by zero and unbounded above, which makes it lopsided. Logging fixes that and makes the two crises more comparable to each other: raw skewness was 1.31 for 2008 and 3.46 for COVID, and after logging, −0.36 and 0.91.

4.3 Predictors

Four traits, all measured **strictly before** each crisis (2007 for the 2008 crisis, 2019 for COVID):

- **Self-employment** — proprietor share of income (BEA). We had no strong prior here. Self-employment could speed recovery, because small operators adapt quickly, or slow it, because they have thinner buffers.
- **Migration** — net migration rate (IRS). A proxy for economic momentum.
- **Pre-crisis unemployment** — baseline unemployment rate (BLS). Our hypothesis was that already-struggling labor markets would recover worse.
- **Working-age share** — percent of the population of working age (PEP).

4.4 Controls

Income, poverty rate, log population, log density, Rural-Urban Continuum Code, and pre-crisis population trend. These are in every model and we never interpret them. They exist so the four studied predictors are measured fairly.

Two are worth explaining:

Rural-Urban Continuum Code (RUCC) is USDA's 1-to-9 score for how urban a county is. We treat it as an unordered category rather than a number, because the scale is not a straight line — codes 4 through 9 alternate based on whether a county sits next to a metro area, so a one-step move does not mean the same thing everywhere on the scale.

Pre-crisis population trend separates counties that were already shrinking for structural reasons from counties that genuinely got knocked down by a crisis. Without it we would confuse decades-long rural decline with crisis damage.

4.5 No data leakage

Every predictor and control is measured before its crisis year — never during, never after. This rule cost us: we could not use the 2005–2009 American Community Survey for education, because that file contains 2008 and 2009 data. We audited this and confirmed it holds throughout.

5 How We Handle Wealth

This is the methodological core of the project, so it deserves its own section.

Wealth is the strongest predictor of recovery and the least interesting one. We handle it in two ways.

Primary approach: control for it continuously. Income and poverty are in every model. In a multiple regression, a coefficient answers a conditional question. The coefficient on self-employment is not "the effect of self-employment." It is "the effect of self-employment among counties at the same income level." The model builds that estimate by comparing counties that have similar incomes but different self-employment rates.

That is exactly our research question, implemented precisely.

Secondary approach: wealth bands. We split counties into three income groups and re-run the models inside each. Controlling assumes a trait works the same way everywhere on the income scale. Banding lets it work differently. That difference is worth testing rather than assuming, and as we show later, it uncovered a real effect that the pooled model hid.

One technical detail that matters: in our LASSO models we set `penalty.factor = 0` on every control. LASSO normally shrinks weak variables toward zero, and if it shrank income out, our "holding wealth constant" claim would quietly stop being true in the model we were reporting.

6 Methods

We use four methods. Running several and checking whether they agree is stronger evidence than trusting any one of them.

Multiple linear regression (OLS). The baseline. Gives us a coefficient and a p-value for each predictor. We use HC1 robust standard errors because our diagnostics showed the migration variable's spread widening in small-population counties — a rate computed on a small denominator swings wildly. Robust errors fix the resulting inference problem without altering the data.

LASSO. A regression that penalizes complexity. It shrinks weak coefficients toward zero and drops the weakest entirely, which makes it a variable-selection tool. Cross-validated over 10 folds.

Relaxed LASSO. Standard LASSO does two jobs at once: picking variables and shrinking them. The shrinking makes surviving coefficients look smaller than they really are. Relaxed LASSO separates the jobs — it picks the same variables, then refits without the shrinkage, so coefficients come back at full size.

Random forest. Builds hundreds of decision trees on random subsets of the data and averages them. It captures curved and interacting relationships that a linear model misses, and it reports which variables matter most for prediction. We run it twice per crisis: once **with** income and poverty, to get the best possible prediction accuracy, and once **without**, to get variable importance. Income dominates the importance rankings if left in, which would drown out the four traits we actually want to compare.

Every model uses the same seed, the same predictor set, and the same outcome transform for both crises. That symmetry is what makes the cross-crisis comparison meaningful.

7 Results

7.1 What recovery actually looked like

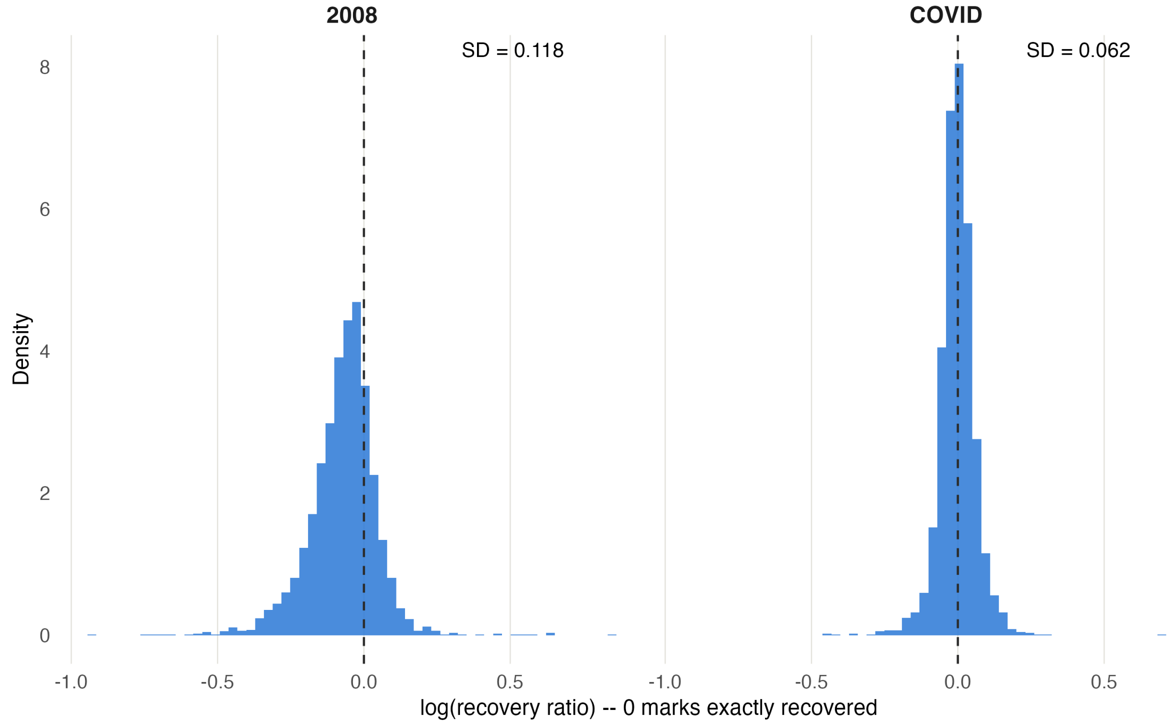

Figure 1: Distribution of log recovery ratio for both crises. The vertical line marks zero, which is the point where a county returned exactly to its pre-crisis employment peak.

The two crises produced very different distributions.

Table 2: Outcome distributions on the log scale

	2008	COVID
Counties in model	3,073	3,017
Mean log recovery	−0.072	−0.002
SD of log recovery	0.118	0.062

The average county was about 7% below its pre-crisis peak at the 2008 checkpoint, and essentially exactly at peak at the COVID checkpoint. More importantly, **2008 spread counties nearly twice as far apart.** Hold onto that — it matters for how we read the prediction results later.

7.2 The main result

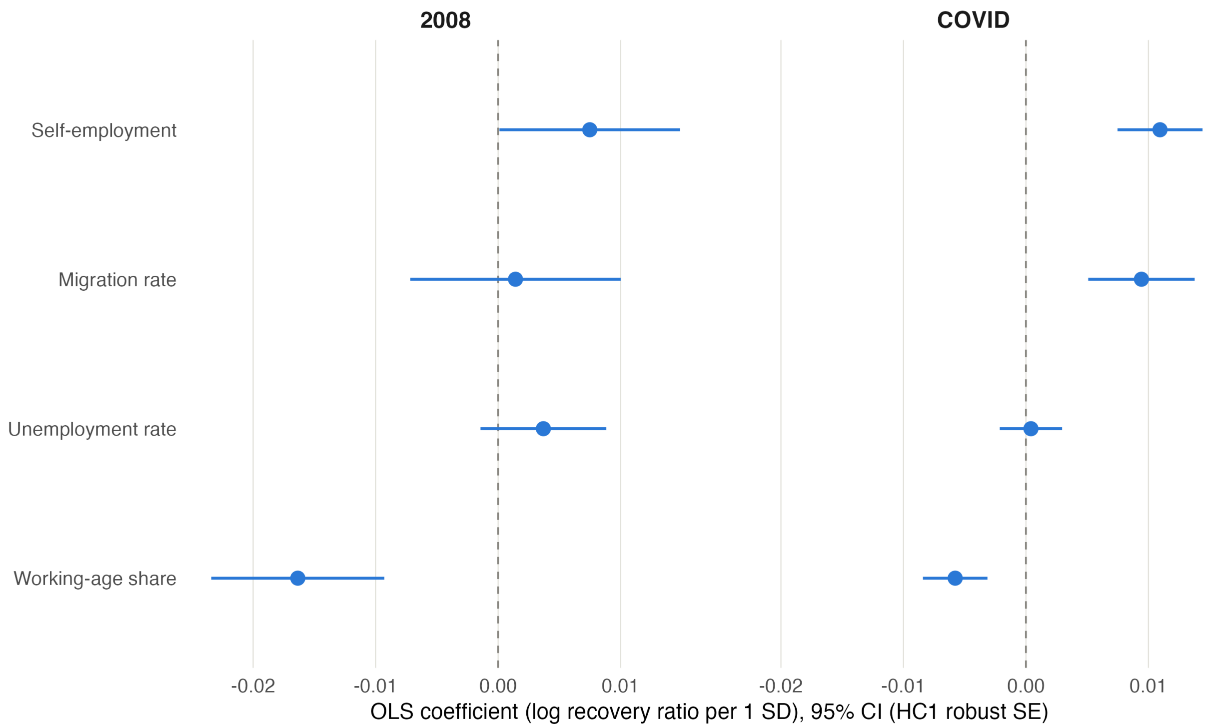

Figure 2: OLS coefficients for the four studied predictors, with 95% confidence intervals from robust standard errors. Predictors are standardized, so each coefficient is the effect of a one-standard-deviation increase.

Table 3: Main OLS results. Bold indicates significance at the 0.05 level.

Predictor	2008	COVID
Working-age share	**−0.0164** (p<.001)	**−0.0058** (p<.001)
Self-employment	**+0.0075** (p=.047)	**+0.0109** (p<.001)
Migration rate	+0.0014 (p=.75)	**+0.0094** (p<.001)
Unemployment rate	+0.0037 (p=.16)	+0.0004 (p=.75)

Four findings, one per row.

Working-age share predicts recovery in both crises — negatively. This was our largest 2008 effect and it runs opposite to intuition. Counties with *more* working-age people recovered worse. A plausible explanation is that retirement-heavy counties have income streams — Social Security, pensions — that do not depend on local employment. That money keeps flowing into local businesses during a downturn and props up demand. We cannot test that mechanism with our data, so we offer it as a hypothesis rather than a conclusion.

Self-employment predicts recovery in both crises — positively. The adaptability story wins over the fragility story. The effect is larger and more precisely estimated in COVID.

Migration is the sharpest contrast in the project. Statistically invisible in 2008 (p=.75), and one of the strongest predictors in COVID (p<.0001). Not a sign flip — a relevance flip. The same measurement, on the same counties, using the same model, goes from meaning nothing to meaning a great deal depending on which crisis hit.

Pre-crisis unemployment predicts nothing. Our hypothesis was that struggling labor markets would recover worse. Across two crises, four methods, and 3,000 counties, we found no support for it in the pooled models. This is a clean null result and we report it as one. (Section 7.6 complicates it in an interesting way.)

7.3 Do the methods agree?

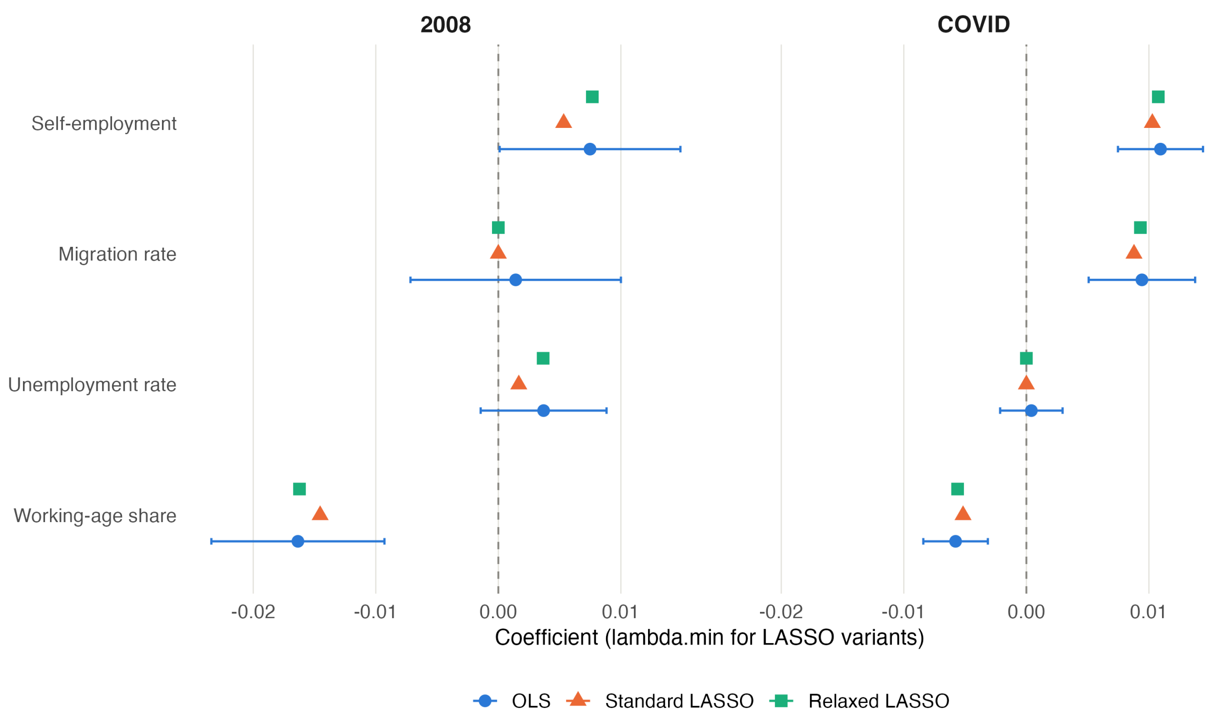

Figure 3: Coefficients from OLS, standard LASSO, and relaxed LASSO side by side. Where the three points stack on top of each other, the methods agree.

They do, and this is one of the more reassuring results in the project.

Table 4: Three methods, same coefficients

Predictor	Crisis	Standard LASSO	Relaxed LASSO	OLS
Self-employment	2008	0.00533	0.00767	0.00748
Unemployment	2008	0.00167	0.00366	0.00369
Working-age share	2008	−0.01454	−0.01623	−0.01635
Self-employment	COVID	0.01028	0.01076	0.01094
Migration	COVID	0.00878	0.00930	0.00943

Predictor	Crisis	Standard LASSO	Relaxed LASSO	OLS
Working-age share	COVID	−0.00517	−0.00561	−0.00578

Read across any row. Standard LASSO produces the smallest number, because it is shrinking. Relaxed LASSO removes the shrinkage and lands almost exactly on the OLS value. Two of the 2008 estimates match OLS to four decimal places.

This tells us our OLS coefficients are not an artifact of one method's assumptions. Three approaches with different mathematics arrive at the same magnitudes.

Selection agrees too. For COVID, LASSO keeps exactly the three predictors that OLS found significant. For 2008 there is one mismatch — LASSO keeps unemployment, which OLS did not find significant — and Section 7.6 explains why.

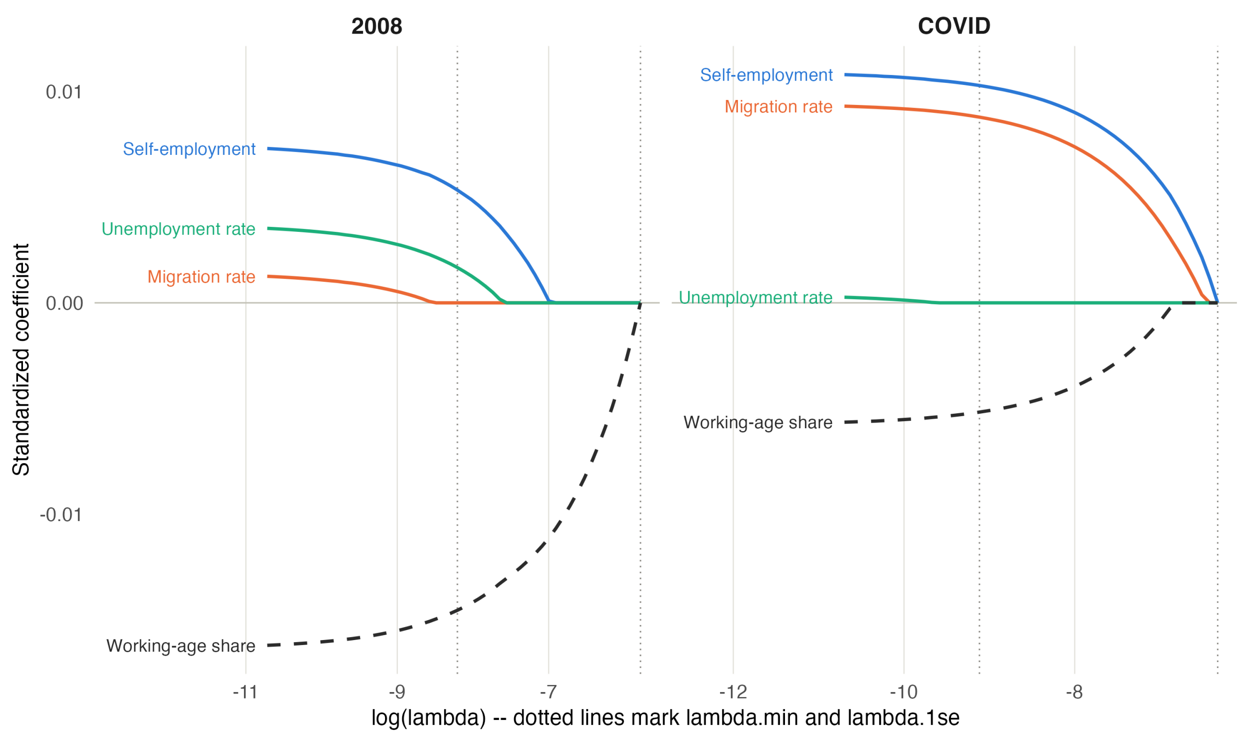

Figure 4: LASSO coefficient paths. As the penalty on the horizontal axis increases, coefficients are squeezed toward zero. Vertical lines mark the two standard cross-validated penalty choices.

One honest note about LASSO. At the more conservative penalty setting (`lambda.1se`), all four studied predictors drop to zero in both crises. This is not mysterious. Our controls are unpenalized, so the model already has income, poverty, population, density, population trend, and eight RUCC categories doing work before the studied predictors get a chance. Against that baseline, dropping four small coefficients costs almost nothing in prediction error.

The correct reading is that these predictors add little **predictive** value beyond the controls — which is entirely compatible with them being statistically detectable and real. We report the less conservative setting as our primary LASSO result and let OLS and random forest carry the argument.

7.4 Which predictors matter most?

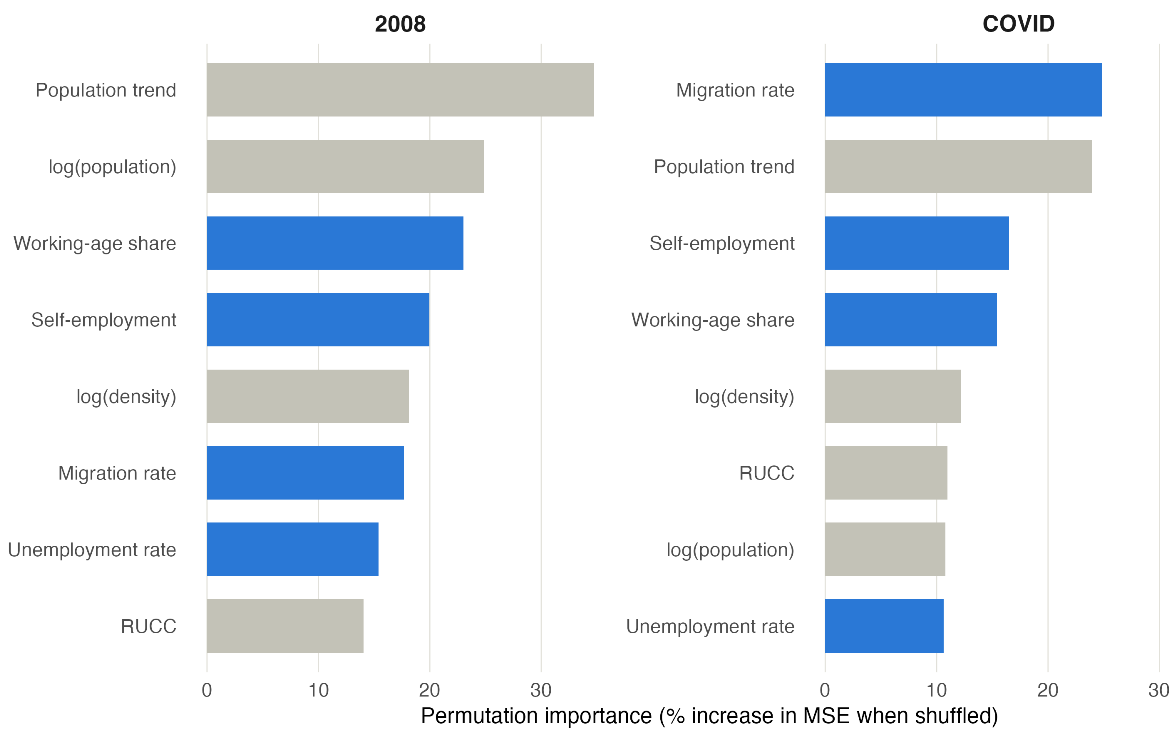

Figure 5: Random forest permutation importance, from the models run without income and poverty so the four studied traits compete against each other rather than against wealth.

The random forest rankings tell the same story as the regression, which is worth noting because it is a completely different kind of model.

Migration ranks **first for COVID and sixth for 2008** — the same relevance flip the regression found. Unemployment ranks near the bottom in both crises. The methods disagree about nothing important.

7.5 How predictable is recovery? (And the test that changed our answer)

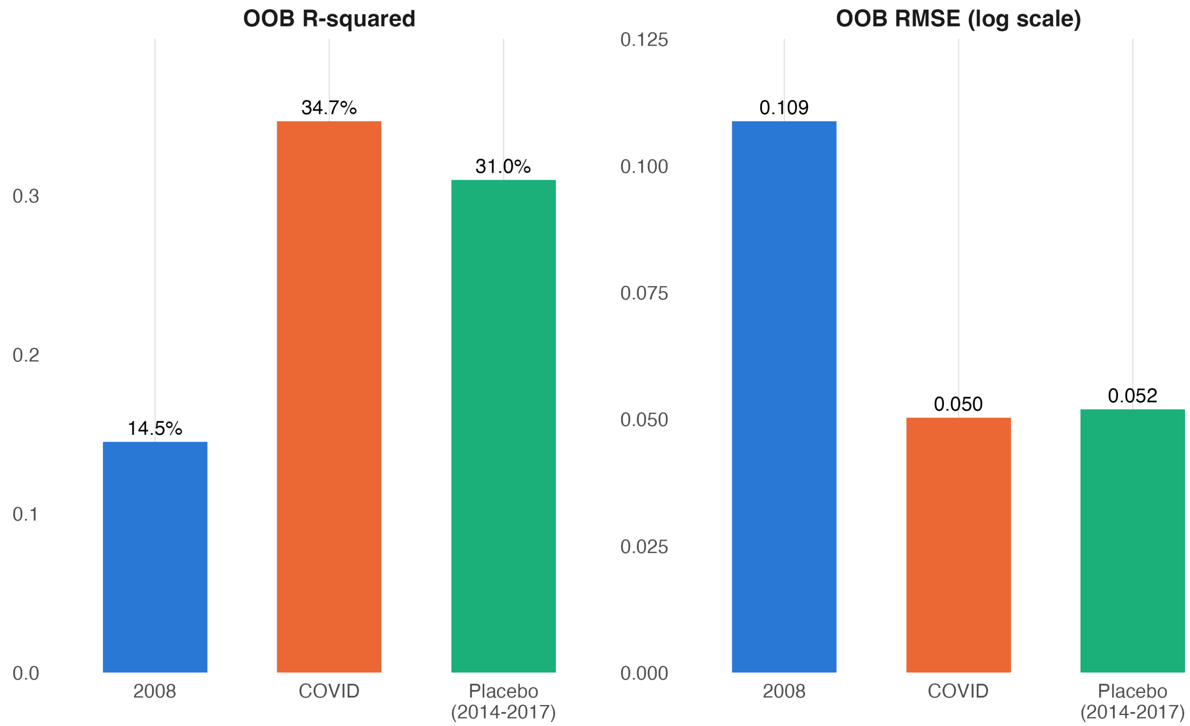

Figure 6: Out-of-bag R-squared and RMSE for the two crises and for a calm-period placebo test.

We originally planned to headline this comparison:

	OOB R^2	OOB RMSE
2008	14.5%	0.109
COVID	34.7%	0.050

COVID recovery looked more than twice as predictable as 2008 recovery. Our interpretation was that COVID hit through local economic structure we could measure, while 2008 arrived through financial channels invisible at the county level.

Then we tested that claim and it did not survive.

We ran a placebo: same predictors, same controls, same methods, but predicting employment change over 2014–2017, an ordinary non-crisis stretch. If COVID's predictability were really about COVID, a calm period should look very different.

	OOB R^2	OOB RMSE
2008	14.5%	0.109
Placebo (2014–17)	**31.0%**	**0.052**
COVID	34.7%	0.050

The calm window came back at 31%, sitting almost on top of COVID's 34.7%. So there is nothing special about COVID's predictability. County employment is just generally about 31% predictable from these traits, and COVID recovery behaved like a normal period.

But the test did not destroy the finding — it relocated it. Now we have a baseline, and against that baseline:

> A global pandemic hit U.S. counties along completely ordinary lines. The 2008 financial crisis was only **half** as predictable as a normal year.

2008 is the anomaly, not COVID. That is a more surprising claim than the one we started with, and it survives its own robustness test. Most people asked to guess would say the pandemic was the more chaotic, less predictable event. The data says the opposite.

One more piece of supporting evidence. 2008 had *more* variation in outcomes (SD 0.118 vs 0.062) and *less* explanation of it. More dispersion combined with worse predictability is exactly the signature of a shock imposed from outside a local economy rather than arising from its structure.

Two limitations on the placebo. The calm window is three years rather than four to five, and it falls inside a growth period. Its population-trend control also uses a three-year window instead of the seven-year window in our main models, because the Census changed its population estimation methodology between those periods and mixing the two would produce a meaningless number.

7.6 Does wealth change the story?

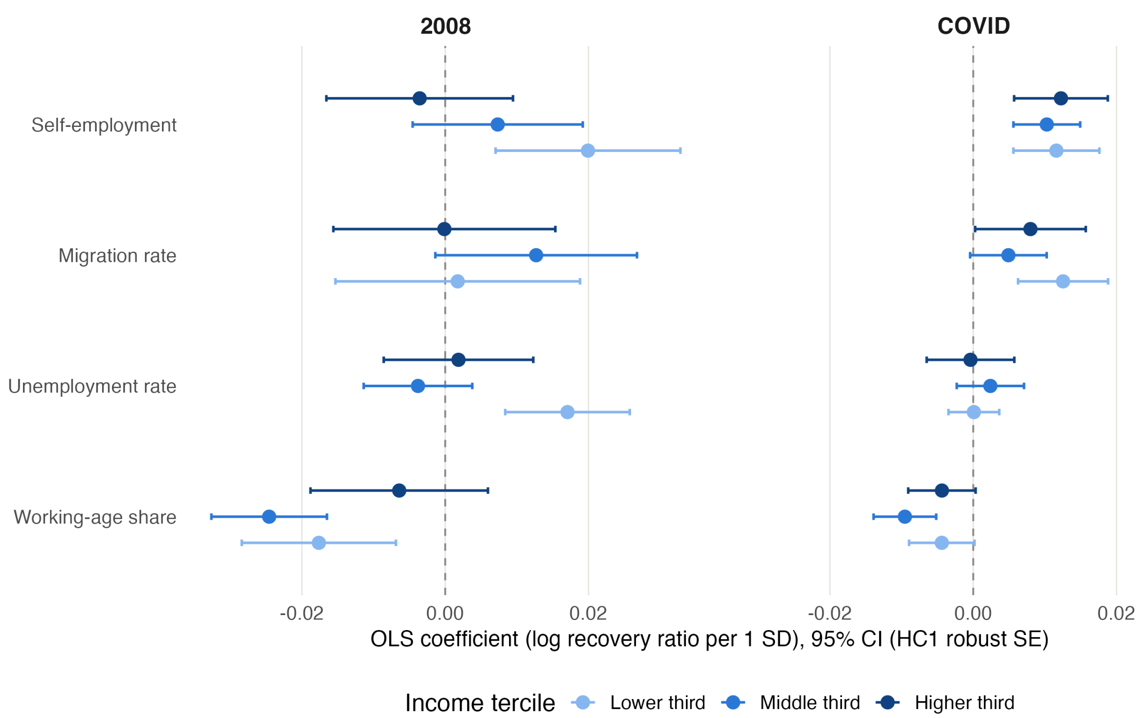

Figure 7: OLS coefficients estimated separately within each of three income bands, for both crises.

This is where banding earned its place. The pooled models were hiding something.

Unemployment: null overall, strongly significant among poor counties. Pooled 2008 p=.16. In the lower-income band alone: coefficient +0.0171, **p=.0001**.

Note the direction. **Positive.** Higher pre-crisis unemployment associated with *better* recovery in poor counties — the opposite of our hypothesis. The likely explanation is mean reversion: a county sitting at 9% unemployment has slack in its labor market and room to reabsorb workers. A county at 3% is already near full employment and has nowhere to climb.

This also explains why LASSO kept unemployment for 2008 while OLS called it insignificant. There is a real effect there; it just only exists in part of the income distribution, so pooling averages it away.

Self-employment changes sign across bands in 2008.

Income band	2008 coefficient
Lower	+0.0199
Middle	+0.0073
Higher	−0.0036

Self-employment helped poorer counties recover from the financial crisis and slightly hurt richer ones. Both of our competing hypotheses were right, in different places.

COVID shows no such pattern. All four predictors hold the same sign across all three bands. Whatever COVID did to county economies, it did it fairly uniformly across the wealth distribution, while 2008's effects depended heavily on how wealthy a county already was.

That is itself a cross-crisis difference, and it is one we would have missed entirely if we had only run the pooled models.

8 Extension: Adding Education and Industry

Our main analysis uses four predictors because those are the only ones available for all 3,000 counties before 2008. We tested a wider set on a smaller sample.

8.1 Why the sample shrinks

Education and industry data for 2007 come from the American Community Survey, and the ACS only publishes for counties above a population threshold. The best pre-crisis file — the 2005–2007 three-year estimates — covers counties above 20,000 people. That drops us to **1,766 counties.**

Those are not a random 1,766. The threshold removes small rural counties specifically. So this extension is not a smaller version of the main analysis; it is an analysis of **larger counties only.** We report it as such.

8.2 Reading the extension honestly

Comparing the main analysis directly to the extension would be misleading, because two things change at once: the sample shrinks *and* new predictors enter. So we built a middle model — the original four predictors, on the smaller sample — that lets us change one thing at a time.

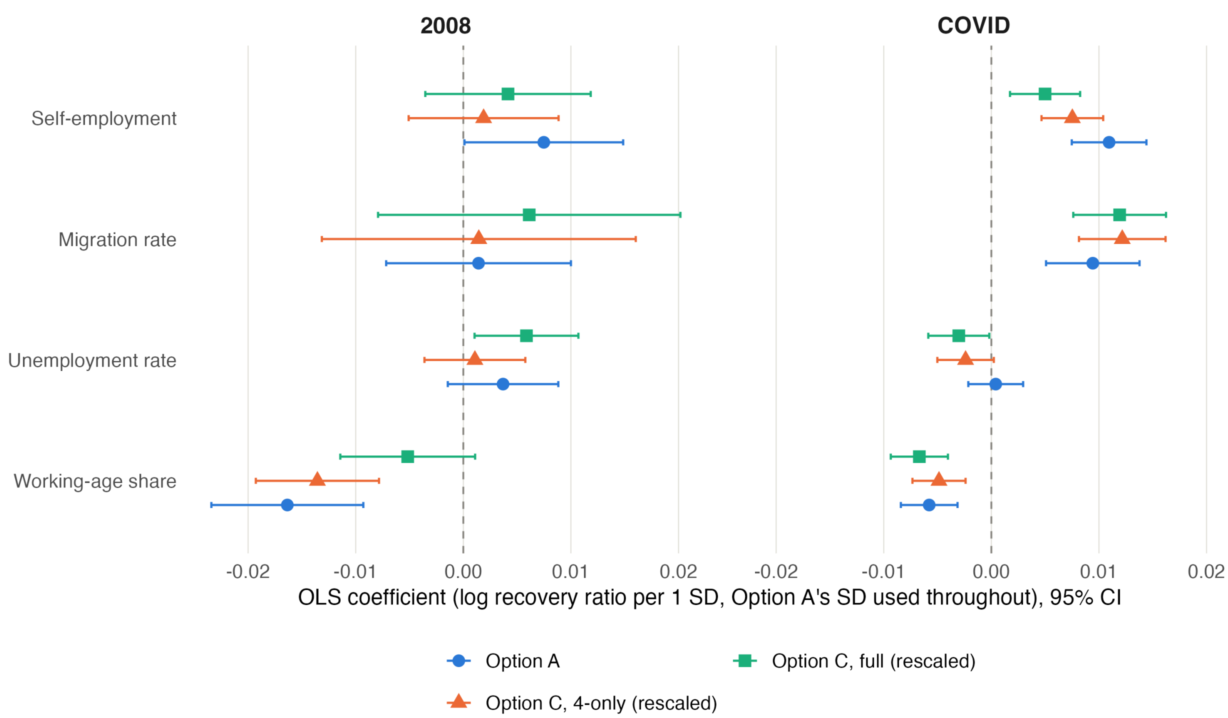

Figure 8: The four original predictors across three specifications: the full-sample main analysis, the same four predictors on the reduced sample, and the full extension model. Reading left to right isolates the effect of the sample change and then the effect of the added predictors.

All coefficients here are standardized on the **main analysis** scale, not the extension's own scale. This matters more than it sounds. A standardized coefficient depends on how much the predictor varies in the sample, and self-employment varies less once rural counties are removed — so part of any apparent shrinkage would be a measurement artifact rather than a real change. Rescaling removes that artifact.

After rescaling, here is what actually changes:

Caused by the smaller sample: self-employment loses significance in 2008 ($p=.047 \rightarrow p=.595$). This makes sense. Self-employment is concentrated in small rural counties, and those are exactly the counties the threshold removed. The effect is real; it just lives in the places this sample excludes.

Caused by the added predictors: working-age share loses 2008 significance ($p<.0001 \rightarrow p=.106$), and unemployment becomes significant in both crises with opposite signs. That last one is instability, not discovery.

Unaffected by either: migration. Null in 2008, significant in COVID, throughout.

8.3 The industry finding

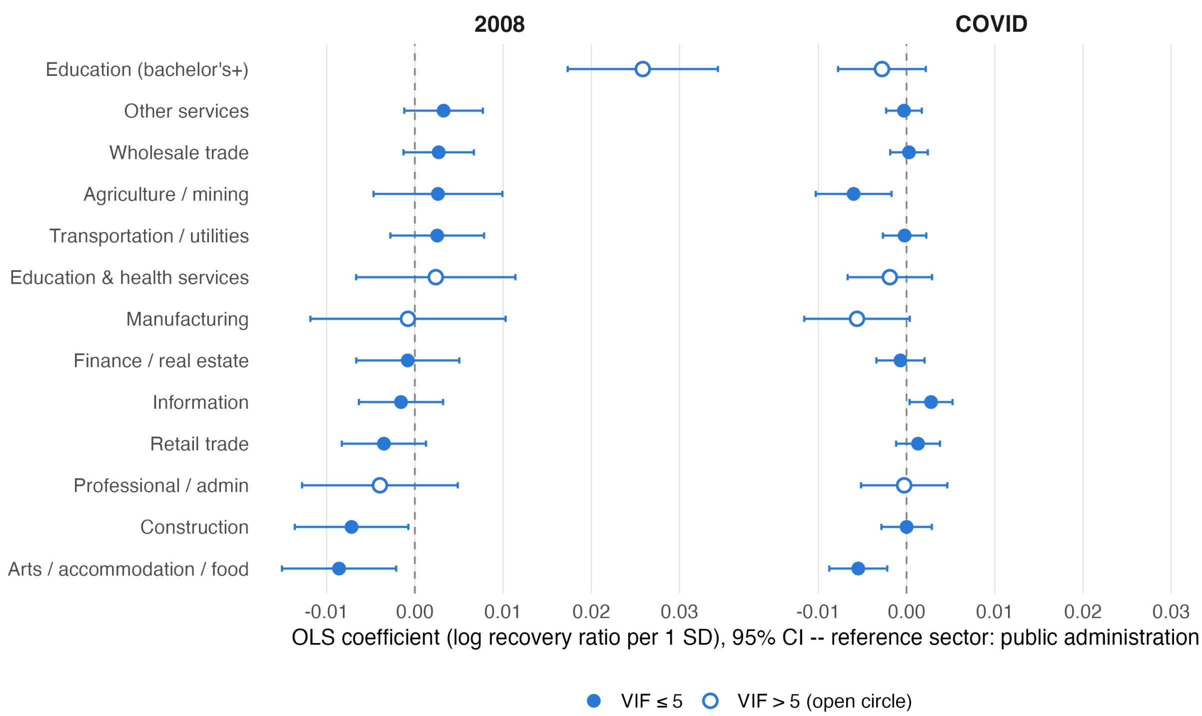

Figure 9: Coefficients for education and the twelve industry sectors. Public administration is the omitted reference category. Open markers indicate variables with variance inflation above 5, whose individual estimates should be treated cautiously.

This is what the extension was for.

Table 5: Significant industry sectors. Bold indicates significance at the 0.05 level.

Sector	2008	COVID
Construction	**−0.0072** (p=.029)	0.0000 (p=.995)
Information	−0.0015 (p=.547)	**+0.0028** (p=.025)
Agriculture / mining	+0.0027 (p=.460)	**−0.0060** (p=.006)
Arts / accommodation / food	**−0.0085** (p=.010)	**−0.0055** (p=.001)

Note that information and agriculture/mining do not merely weaken in 2008 — they also point the opposite direction from their COVID estimates, though neither 2008 estimate is distinguishable from zero.

Construction is the cleanest result in the entire project. Significant and negative in 2008, and in COVID its p-value is **0.995** — as close to zero effect as a coefficient can get. A crisis that started in housing hurt counties whose workers built houses. A crisis that started with a virus did not touch them at all.

That is the mechanism our research question was looking for. It is not just that different traits predicted recovery from different crises — it is that the reason is legible. Same trait, opposite relevance, depending on how the shock arrived.

Information runs the other way: irrelevant in 2008, significant and positive in COVID, consistent with remote-work-capable sectors weathering shutdowns.

One result contradicts the obvious story. Arts, accommodation, and food service — the sector everyone associates with COVID lockdowns — is significantly negative in *both* crises, and the effect is actually **larger in 2008**. We do not have a good explanation for this and we are not going to invent one. It is worth flagging as something the data says that our intuitions did not predict.

8.4 Race and ethnicity

The extension also includes county racial and ethnic composition, entered as population shares with white share as the omitted reference category. Several are statistically significant.

Table 6: Race and ethnicity coefficients. White share is the omitted reference.

Predictor	2008	COVID
Black share	**+0.0082** (p=.0033)	+0.0023 (p=.071)
Hispanic share	**+0.0154** (p<.0001)	**+0.0051** (p=.0002)
Asian share	**+0.0037** (p=.029)	−0.0015 (p=.129)
Other race share	+0.0008 (p=.784)	**+0.0039** (p=.0051)

Hispanic share is significant in both crises, and its 2008 coefficient is among the largest in the entire project — comparable in magnitude to working-age share. All the significant estimates are positive, meaning that counties with larger non-white population shares tended to show higher recovery ratios than otherwise similar counties with larger white shares.

We are reporting these because we ran them, not because we can explain them. Four things constrain how far this can be read:

This is an ecological association. Our unit of analysis is the county, not the person. A county-level relationship between racial composition and recovery says nothing about outcomes for individuals of any race. Treating it otherwise is the ecological fallacy, and it is one of the easiest mistakes to make with data shaped like ours.

Racial composition is bound up with geography. Counties with large Hispanic populations are concentrated in the Southwest, Texas, and Florida. Counties with large Black populations are concentrated in the South. Regional economic trajectories over these periods differed substantially for reasons that have nothing to do with the composition variable itself, and our controls include no region term.

Recovery is a ratio, not a level. A county recovering to 100% of its pre-crisis peak is not the same as a county being economically well-off. Places starting from lower employment bases can post higher recovery ratios while remaining worse off in absolute terms. Our outcome measures the bounce, not the destination.

The mechanism is unknown. We can construct plausible stories — differences in industry mix that our twelve sectors do not fully capture, differences in population growth, differences in labor force reentry — but we tested none of them. Offering an untested mechanism for a result this sensitive would be worse than offering none.

These variables were included so that the other extension predictors would be estimated conditional on them, which is a defensible reason to have them in the model. We do not treat them as a finding, and we would not draw any policy conclusion from them without a design built specifically to study this question.

8.5 What we are not claiming

Education is not a clean finding. It produced the largest single coefficient in the extension (+0.0260 for 2008, $p<.0001$), and we are deliberately not building on it. Education correlates with income at $\mathbf{r = 0.68}$ in 2007 and $\mathbf{r = 0.73}$ in 2019, with variance inflation of 6.0 and 8.5. Roughly half its variance is income. In a project designed specifically to hold wealth constant, a predictor that is half wealth cannot be presented as a distinct finding.

The extension does not add explanatory power. Adding 17 predictors moved R^2 by 2.8 points in 2008 and 0.7 points in COVID. Its value is the industry contrast, not better prediction.

Seven variables in the full extension model have variance inflation above 5. Education, manufacturing, professional services, education/health, income, poverty, and density. Their individual coefficients are unstable and we do not interpret them one by one. Construction and information — our two actual findings — are both below that threshold.

8.6 An unplanned finding

Comparing the main analysis to the four-predictor model on the reduced sample turned up something we were not looking for:

	Full sample (~3,000)	Counties over 20,000 (1,766)
2008 OOB R^2	14.5%	26.6%
COVID OOB R^2	34.7%	55.8%

Predictability roughly **doubles in both crises** simply from dropping counties under 20,000 people. Same predictors, same methods, same everything else.

This is our small-county measurement noise, quantified. BLS employment series for small counties are built on small samples, so peak and trough detection there picks up sampling noise alongside real economic movement. We had been listing this as a limitation based on reasoning. Now it is a number.

Also worth noting: the 2008-versus-COVID predictability gap survives the sample change. It is 2.39x on the full sample and 2.10x on the restricted one.

9 Can the Last Crisis Predict the Next One?

We saved this test for last because it asks the question a policymaker would actually care about: if you had studied 2008 carefully, would you have been ready for 2020?

We trained a random forest on 2008 data only, then handed it 2019 predictor values and asked it to predict COVID recovery. Then we did the reverse.

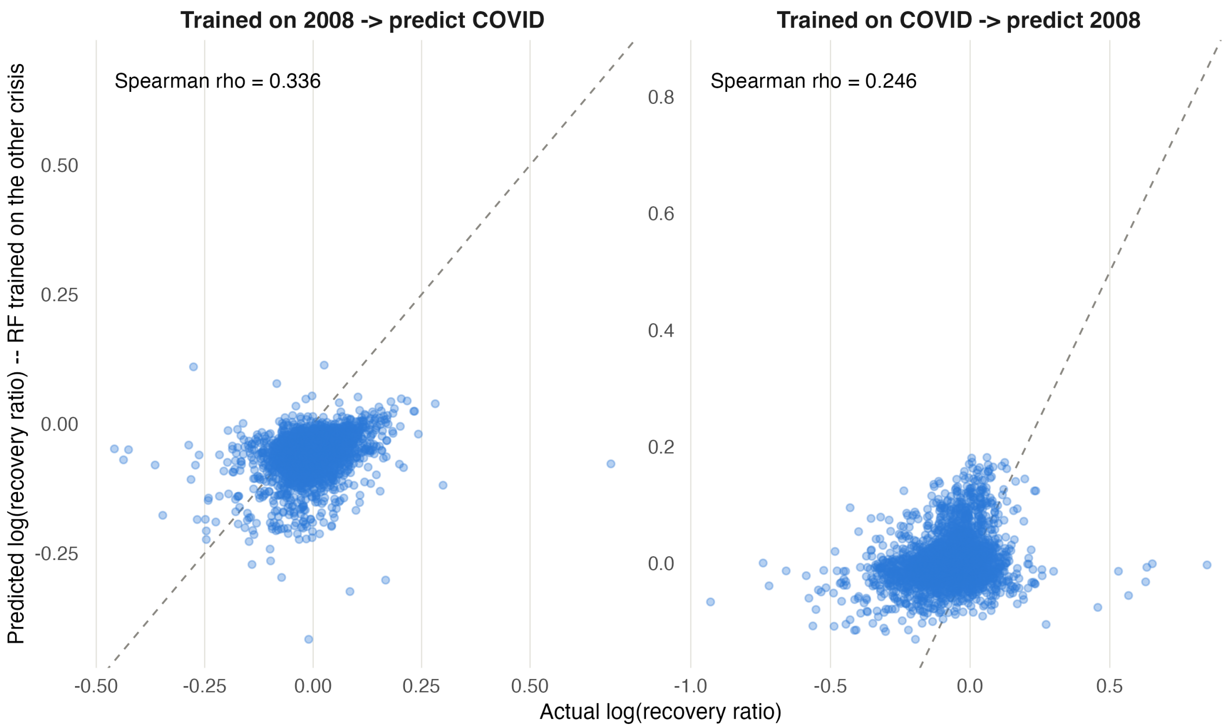

Figure 10: Predicted versus actual recovery for both cross-crisis directions. The diagonal line marks perfect prediction.

Direction	R^2	RMSE	Spearman
Train 2008 → predict COVID	−0.801	0.0835	0.337
Ceiling: COVID → COVID	*0.354*	*0.0500*	*0.587*
Train COVID → predict 2008	−0.386	0.1341	0.246
Ceiling: 2008 → 2008	*0.155*	*0.1048*	*0.413*

Ignore the negative R^2 values. They are mechanical. The two crises sit at different levels — 2008 counties averaged 7% below peak, COVID counties averaged right at peak — so a 2008-trained model predicts everyone too low for COVID and scores worse than simply guessing the average. This tells us nothing interesting.

Spearman correlation is the fair test. It asks whether the model can correctly *rank* counties, ignoring the level difference. Against each crisis's own ceiling, the cross-crisis models retain about 57% and 60% of their ranking ability.

So the answer is: **partially.** Studying the last crisis gets you a little over half of what you would want to know about the next one. Some resilience is durable — a property of the place. A substantial share is not — it depends on how the shock arrives.

That is the project's answer to its own question, stated as precisely as our data allows.

10 Robustness

We ran three checks on the main analysis. All three held.

Excluding small-employment counties. Counties flagged for very small employment bases are the most likely to have noisy peak and trough detection. We removed the union of counties flagged in either crisis — the same 123 counties dropped from both models, so the samples stay comparable — and re-ran. Every studied coefficient kept its sign, and magnitudes moved only slightly (2008 self-employment went from 0.0075 to 0.0064).

Winsorizing the outcome. We capped the recovery ratio at the 1st and 99th percentiles to check whether extreme counties were driving results. Self-employment in 2008 moved from 0.0075 to 0.0066. Nothing else changed qualitatively.

Method agreement. Covered in Section 7.3. OLS, LASSO, relaxed LASSO, and random forest agree on which predictors matter and, where comparable, on their magnitudes.

We also documented an earlier specification that failed. We tested a version with education and industry data drawn from single-year ACS and County Business Patterns, which collapsed the sample to roughly 735 counties and broke the wealth bands. We kept it as a documented branch rather than deleting it, because knowing which specifications fail is part of knowing why the final one was chosen. Notably, that version found no cross-crisis industry flip — the finding only appeared once we switched to a data source without confidentiality suppression. Data quality changed the answer.

11 Limitations

This is associational, not causal. We observe counties as they are. We cannot rule out that some unmeasured factor drives both a county's traits and its recovery.

Two crises is two data points. Our question asks whether traits are consistent *across crises*, and we have exactly two. A trait that behaves differently in 2008 and 2020 might be crisis-specific, or it might be responding to something particular about those two events. We cannot distinguish those with n=2.

Effect sizes are small. Our coefficients are on the order of 0.005 to 0.016 in log units — roughly half a percentage point to 1.6 percentage points of recovery per standard deviation. These are real and they replicate, but they are not large.

Most of recovery is unexplained. Our best model explains about 35% of variation. County recovery depends heavily on things no dataset captures: a single large employer's decisions, local politics, luck.

Small-county employment data is noisy. Quantified in Section 8.5 — dropping counties under 20,000 roughly doubles predictability, which tells us how much noise those counties carry.

We could not use industry data for the full sample. No source publishes pre-2008 county industry data cleanly for all counties. Our industry findings apply only to counties above 20,000 people.

Recovery windows differ in length. COVID's window is shorter and more compressed than 2008's, which was still unfolding years later.

Puerto Rico is excluded — its municipios lack the controls we need, and its pre-2020 baseline was shaped by Hurricane Maria and a debt crisis, making it a fundamentally different case.

12 Conclusion

12.1 What we found

1. **Two traits predicted recovery from both crises.** Working-age share (negatively) and self-employment (positively). Resilience has a durable component.
2. **One trait predicted only COVID recovery.** Migration was statistically invisible in 2008 and among the strongest COVID predictors. Resilience also has a crisis-specific component.
3. **One trait predicted nothing — until we split by wealth.** Pre-crisis unemployment was null in both pooled models, but strongly significant among poor counties in 2008, in the opposite direction from our hypothesis. Pooling hid a real effect.
4. **Industry composition flips between crises.** Construction predicted 2008 recovery and had a p-value of 0.995 in COVID. Information ran the reverse. This is the mechanism behind finding 2.
5. **2008 was the anomaly, not COVID.** A calm baseline period is about 31% predictable. COVID recovery was 34.7% — ordinary. The financial crisis was 14.5% — half as predictable as a normal year.
6. **The last crisis is a partial guide to the next.** A model trained on one crisis retains roughly 57–60% of its ranking ability on the other.

12.2 Why it matters

Federal disaster and recovery aid is allocated by formula, using county characteristics to identify which places need help. Our results say a single formula cannot serve both crisis types. The counties most at risk in a credit crisis are not the counties most at risk in a health crisis, and the pre-crisis traits that would flag them are different.

Our unemployment result sharpens the point. A targeting model built on national averages would have concluded that pre-crisis unemployment carries no information. Inside the lower-income band, it carries a great deal. Anyone allocating aid on pooled national estimates is missing effects that exist in exactly the places aid is meant to reach.

12.3 What we would do next

Add crises. With three or four downturns instead of two, “are these traits consistent” becomes a question we could actually answer rather than illustrate. The 2001 recession is a natural next case: it was concentrated in technology and manufacturing, which makes it a third distinct shock type.

Find full-coverage industry data. Our single cleanest finding — construction — comes from the 1,766-county extension. Getting industry composition for all 3,000 counties would let us test it on the rural counties where the never-recovered pattern is concentrated.

13 Appendix A: Variable Dictionary

Table 7: Main analysis variables

Variable	Definition	Source	Year
`recovery_ratio`	Mean employment in trough year +2 and +3, ÷ peak employment. Peak and trough years are county-specific.	BLS LAUS	county-specific
`proprietor_share`	Proprietors' share of total employment	BEA	2007 / 2019
`migration_rate`	Net migration as share of population	IRS SOI	2007 / 2019
`unemp_rate`	Unemployment rate	BLS LAUS	2007 / 2019
`working_age_share`	Share of population aged 18–64	Census PEP	2007 / 2019
`income`	Median household income	Census SAIPE	2007 / 2019
`poverty_rate`	Share of population below poverty line	Census SAIPE	2007 / 2019
`population`	Total resident population (logged in models)	Census PEP	2007 / 2019
`density`	Population per square mile (logged in models)	Census PEP + land area	2007 / 2019
`rucc`	Rural-Urban Continuum Code, 1–9, as unordered factor	USDA ERS	2003 / 2013
`pop_trend`	Population CAGR, 2000→2007 and 2012→2019	Census PEP	pre-crisis

Extension variables (1,766-county sample only): bachelor's degree share among adults 25 and over (ACS 2005–07 three-year estimates for 2008; ACS 2015–19 five-year for COVID); four race and ethnicity population shares — Black, Hispanic, Asian, and other, with white share as the omitted reference (Census PEP intercensal file for 2007, CC-EST2019 for 2019); and twelve industry sector shares from ACS table C24030, sex by industry for the civilian employed population 16 and over, with public administration as the omitted reference.

All seventeen extension variables are penalized in LASSO; only the six controls and the RUCC dummies carry `penalty.factor = 0`.

14 Appendix B: Technical Notes

Sample. 3,073 counties for 2008, 3,017 for COVID, 3,012 present in both. Puerto Rico excluded. Alaska has eleven FIPS codes appearing in one crisis and not the other, entirely explained by real boundary changes — Wrangell-Petersburg, Skagway-Hoonah-Angoon, and Valdez-Cordova all split during the study period.

Recovery window definitions.

Crisis	Peak search	Trough search	Measurement
2008	2006–2008	peak year +1 through 2013	mean of trough +2 and trough +3
COVID	2018–2019	peak year +1 through 2022	mean of trough +2 and trough +3

Missingness structure. Missing values cluster: six controls plus migration and age structure are missing for the same 92–96 counties, reflecting one shared join gap rather than six independent problems. Self-employment adds 52 more in 2008.

Standardization. Studied predictors are standardized within crisis. Extension models rescale the four original predictors onto the main analysis scale so coefficients are directly comparable across specifications.

Collinearity in the main analysis. Only one pair exceeds $|r| = 0.6$: income and poverty, at -0.75 in 2007 and -0.77 in 2019. Both are controls and this is expected. No studied predictor correlates with any control above 0.6. Migration and population trend sit at 0.58 and 0.57 — related, but under the threshold.

Software. All analysis in R. Key packages: `glmnet` for LASSO and relaxed LASSO, `randomForest`, `estimatr` for robust standard errors, `tidyverse` for data handling, `ggplot2` for figures. Seed set at the top of every script; 10-fold cross-validation; 500 trees per forest.

Reproducibility. Analysis lives in two Quarto documents. Fitted model objects are cached to disk so figures regenerate without re-running models.

One data-sourcing disclosure. The 2017 county employment figures used in the placebo test were retrieved from the Internet Archive's mirror of the BLS file rather than from bls.gov, which blocks automated downloads. Same file, different access point.